

33 — Disaster Recovery Specification

Version 1.0

Product-Level Engineering Specification

Core recovery rule: Restore service safely, preserve evidence, reconcile authoritative state, and never treat restoration alone as permission to create trusted state.

Document	Value
System	CloudShield AI
Specification	33 — Disaster Recovery
Version	1.0
Scope	Product-wide disaster recovery, business continuity, backup, restore, failover, and reconciliation
Primary Cloud	AWS for MVP
Status	Engineering Baseline
Depends On	00–32

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1. Purpose and Scope

This specification defines how CloudShield AI prepares for, responds to, and recovers from failures that materially affect availability, integrity, confidentiality, or the ability to operate the product. It covers application services, workers, databases, object storage, queues, identity/secrets, AWS infrastructure, external dependencies, AI services, DAIL, evidence, observability, and trusted-state reconciliation.

Disaster recovery is not limited to restoring compute. A successful recovery must establish that CloudShield AI can resume safe operation, account for incomplete or duplicated work, reconcile external cloud state, preserve audit/evidence integrity, and re-establish normal lifecycle controls.

Recovery restores capability; verification and governance determine whether recovered state can again become trusted.

1.1 In Scope

- Business continuity and disaster recovery objectives.
- Backup, restore, failover, replication, and recovery architecture.
- Recovery of application, database, object storage, queues, workers, identity, secrets, keys, and infrastructure.
- AWS region/AZ/service failure scenarios and customer-cloud integration failure.
- AI/LLM provider outages and degraded intelligence operation.
- DAIL recovery and lifecycle integrity.
- Data reconciliation, idempotency, duplicate work, ambiguous external outcomes, and trusted-state restoration.
- Recovery testing, drills, runbooks, communications, evidence, and operational metrics.

1.2 Out of Scope

- Replacing the AWS/Terraform implementation details in Specification 27.
- Replacing security incident response in Specification 32.
- Defining a full enterprise business continuity program outside CloudShield AI.
- Guaranteeing zero data loss for every failure class; recovery targets are explicitly defined by workload and criticality.

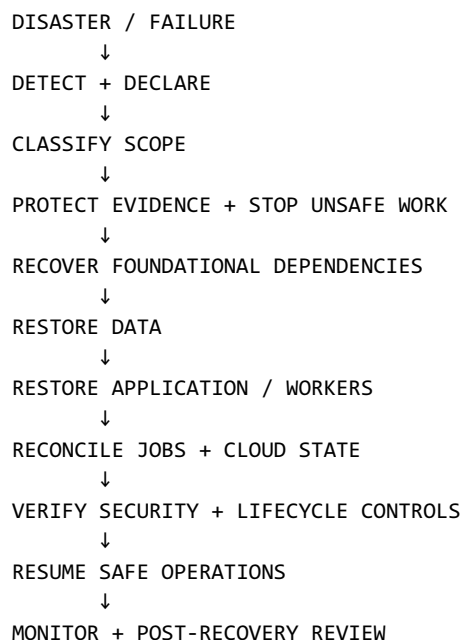
2. Disaster Recovery Objectives

Objective	Requirement	Success Condition
Availability	Restore critical product capability within defined recovery objectives.	Users can safely access and operate supported critical workflows.
Durability	Protect authoritative product data and evidence.	Recoverable copies exist and restore successfully.
Integrity	Avoid silently restoring inconsistent state.	Recovered state passes integrity and reconciliation checks.
Continuity	Preserve safe degraded operation where full service is unavailable.	Read-only/limited workflows remain available when designed.
Reconciliation	Resolve incomplete distributed operations.	Jobs, cloud mutations, lifecycle state, and evidence converge.
Security	Recovery does not weaken security boundaries.	Least privilege, tenant isolation, audit, and safety controls remain active.
Governance	Recovery does not bypass lifecycle authority.	Trusted state is re-earned after recovery where necessary.

2.1 Recovery Principles

- Recover in dependency order rather than simply restarting all services.
- Restore data before enabling mutation-heavy workflows.
- Prefer deterministic recovery procedures over ad hoc operator actions.
- Preserve original evidence and audit history.
- Treat unknown or ambiguous state as unresolved until authoritative reconciliation.
- Fail closed on missing security, authorization, or governance prerequisites.
- Keep recovery automation least-privileged and reversible where practical.
- Test restoration regularly; an untested backup is not a recovery control.
- Do not assume a recovered application database represents the current cloud state.

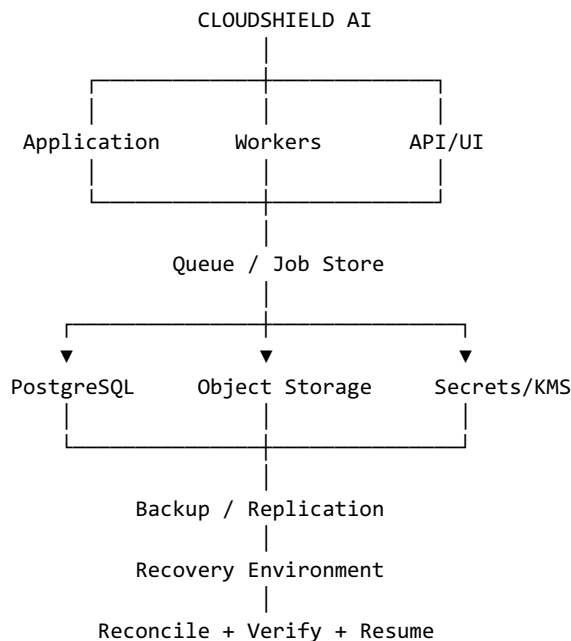
3. Business Continuity Model



3.1 Recovery Modes

Mode	Purpose	Examples
Normal Recovery	Recover from localized failure without changing operating region.	Worker crash, database instance failure.
Degraded Operation	Continue limited safe functionality while a dependency is unavailable.	AI unavailable; read-only security views remain available.
Failover	Move service capability to alternate infrastructure.	AZ/region/service failure.
Restore	Reconstruct service/data from backups or snapshots.	Corrupt database, deleted object store.
Reconciliation	Resolve state divergence after distributed failure.	Cloud mutation ambiguity, duplicate jobs.
Emergency Isolation	Prevent further damage during a destructive/security event.	Compromise, ransomware, integrity incident.

4. Recovery Architecture



Recovery architecture should maintain independent recoverability of critical data classes. Application compute is treated as replaceable; authoritative data, evidence, configuration, and recovery metadata require durable protection.

5. Failure and Disaster Taxonomy

Class	Examples	Primary Recovery Concern
Compute	Container/instance/task failure, deployment failure.	Replace capacity and resume jobs.
Availability Zone	AZ networking or service impairment.	Reschedule/fail over within region.
Region	Regional AWS outage or severe regional degradation.	Restore/fail over to secondary region if configured.
Data	Database corruption, accidental deletion, inconsistent state.	Point-in-time restore and reconciliation.
Storage	Object loss/corruption or access failure.	Restore protected artifacts/evidence.
Security	Compromise, ransomware, malicious deletion.	Isolate, preserve evidence, restore known-good state.

Identity	IAM/IdP outage or credential compromise.	Recover controlled access without broadening privileges.
Dependency	LLM, SIEM, notification, ticketing, external integration outage.	Degrade safely or queue work.
Human/Process	Incorrect deployment, destructive operator action, bad configuration.	Rollback, restore, audit, corrective controls.

6. Recovery Tiers and Criticality

Tier	Workload	Recovery Priority
Tier 0	Identity/security controls, authorization, audit/evidence durability, recovery metadata.	Highest; restore foundational control and data integrity first.
Tier 1	Core API, tenant isolation, asset inventory, findings/risk, lifecycle state.	Immediate product restoration.
Tier 2	Discovery/security workers, dependency/impact, DAIL verification, reporting.	Restore after core state is safe.
Tier 3	AI enrichment, external integrations, analytics, experiments, non-critical notifications.	Restore or re-enable after critical path.

Tiering is a recovery priority model, not permission to bypass security controls.

7. Recovery Time and Recovery Point Objectives

Final RTO/RPO values must be validated through infrastructure sizing, AWS service capabilities, cost constraints, and business requirements. The following are engineering baseline targets for MVP planning rather than guarantees.

Capability	MVP Target RTO	MVP Target RPO	Notes
Critical API / control plane	≤ 2 hours	≤ 15 minutes	Requires recoverable database and infrastructure automation.
Core database	≤ 1 hour	≤ 15 minutes	Point-in-time recovery expected where supported.
Evidence/object storage	≤ 2 hours	≤ 15 minutes	Durability and integrity are higher priority than immediate write availability.
Async job processing	≤ 2 hours	≤ 15 minutes	Unfinished jobs reconciled before replay.
DAIL state	≤ 2 hours	≤ 15 minutes	Candidate/version/evidence relationships must remain consistent.
AI service	Best effort / degraded	Not authoritative	AI outage must not prevent deterministic security controls.
External integrations	Best effort	Integration-specific	Outbound work can queue; inbound events require deduplication/replay handling.

RPO measures acceptable recoverable data loss. RTO measures target time to restore the capability safely; neither removes the need for post-recovery reconciliation.

8. Dependency Recovery Model

Dependency	Recovery Order	Failure Strategy
Identity / authorization	1	Restore/verify before privileged operations.
Secrets / KMS	2	Restore access paths; rotate compromised credentials if required.

Database	3	Restore authoritative domain state.
Object storage	4	Restore evidence/artifact access and integrity.
Queue / job store	5	Recover durable work state and replay metadata.
Application/API	6	Deploy known-good immutable artifacts.
Workers	7	Resume after idempotency/reconciliation checks.
Cloud integration	8	Revalidate customer-cloud connectivity and permissions.
DAIL	9	Revalidate candidate/verification state before sensitive progression.
AI / external integrations	10	Re-enable progressively after core safety path is healthy.

9. Data Protection and Backup Architecture

- Use automated backups for authoritative relational data with point-in-time recovery where supported.
- Use versioning and integrity protection for security evidence and important object artifacts.
- Back up recovery-critical configuration and infrastructure metadata without storing unnecessary secrets in backup artifacts.
- Maintain backup copies in a failure domain distinct from the primary workload where practical.
- Protect backup access with separate least-privileged identities and strong audit controls.
- Encrypt backups at rest and in transit.
- Define retention classes for operational data, security evidence, audit data, and recovery metadata.
- Document what is reconstructed from source systems versus what must be restored from backup.

9.1 Backup Classes

Class	Examples	Protection
Authoritative Domain	Tenants, assets, findings, candidates, lifecycle state, approvals.	Database backup/PITR + integrity validation.
Evidence	Security evidence, verification artifacts, reports.	Versioned/immutable object storage + replication.
Operational State	Jobs, idempotency records, event offsets.	Durable database/queue state.
Configuration	Policy versions, detection versions, integration metadata.	Version-controlled/config backup.
Infrastructure	Terraform state, deployment metadata, artifact references.	Protected remote state + artifact registry.
Secrets	Secret values/keys.	Managed secret/KMS recovery mechanisms; never plaintext backup.

10. Database Recovery

1. Declare the recovery event and freeze unsafe writes if integrity is uncertain.
2. Identify the latest known-good recovery point.
3. Restore into an isolated recovery environment where practical.
4. Run schema/version compatibility and integrity checks.
5. Validate tenant isolation, foreign-key/domain invariants, lifecycle invariants, and evidence relationships.
6. Compare recovery metadata against known external observations where available.
7. Promote recovered database access only after recovery approval.

8. Resume application writes gradually and monitor for reconciliation conflicts.

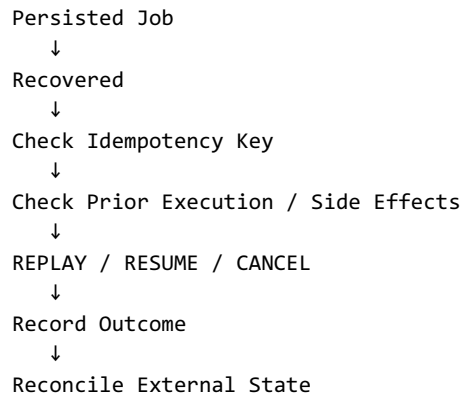
10.1 Database Integrity Checks

- Required tables and indexes exist.
- Referential integrity is valid.
- Candidate versions remain bound to verification and impact records.
- Audit/evidence references are resolvable.
- Tenant ownership and scope boundaries are intact.
- Lifecycle states are valid according to Specification 06.
- No recovered record claims VERIFIED, APPROVED, PROMOTED, or TRUSTED without its required supporting records.

11. Object Storage and Evidence Recovery

- Recover evidence/artifacts independently of application compute.
- Verify object metadata, content hashes, version identifiers, and provenance where recorded.
- Preserve legal/security retention requirements during restore.
- Do not overwrite known-good evidence with uncertain recovered copies.
- Maintain a recovery manifest listing restored objects, source backup/version, integrity status, and validation result.
- Evidence restoration failures affecting security-critical incidents must block closure of those incidents.

12. Queue and Job Recovery



- Every consequential job has a stable job identifier and idempotency strategy.
- Jobs that may have triggered cloud mutations must not be blindly replayed.
- Unknown execution outcome enters reconciliation before retry.
- Duplicate deliveries are deduplicated using durable identifiers.
- Expired or unsafe jobs are quarantined rather than automatically replayed.

13. Application and API Recovery

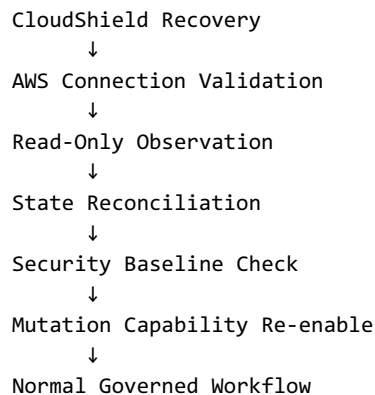
- Deploy immutable, versioned application artifacts from a known-good source.
- Validate database schema compatibility before accepting traffic.
- Restore authentication, authorization, tenant isolation, rate limits, and security middleware before broad exposure.
- Run health, readiness, smoke, security, and lifecycle checks before traffic promotion.
- Use gradual traffic restoration where infrastructure supports it.
- Maintain read-only or degraded mode when mutation prerequisites are unavailable.

14. Worker and Async Processing Recovery

- Workers start only after database, queue, secrets, and authorization dependencies are healthy.
- Recovered workers first reconcile unfinished jobs before accepting broad new work.
- Cloud mutation workers require explicit credential and policy validation after recovery.
- AI workers may resume in advisory mode before mutation-related workflows are re-enabled.
- DAIL workers require integrity validation before verification/promotion workloads resume.
- Worker concurrency should be reduced during initial recovery to avoid thundering-herd effects.

15. AWS and Cloud Integration Recovery

- Revalidate AWS role-assumption permissions after CloudShield recovery.
- Confirm expected account, region, external ID/trust configuration, and permission scope.
- Re-establish read access before mutation access.
- Re-run discovery or targeted observations to determine current authoritative cloud state.
- Treat CloudShield database state as potentially stale until cloud observations are refreshed.
- Do not automatically replay cloud mutations whose prior outcome is uncertain.
- Record provider errors, throttling, eventual-consistency conditions, and reconciliation outcomes.



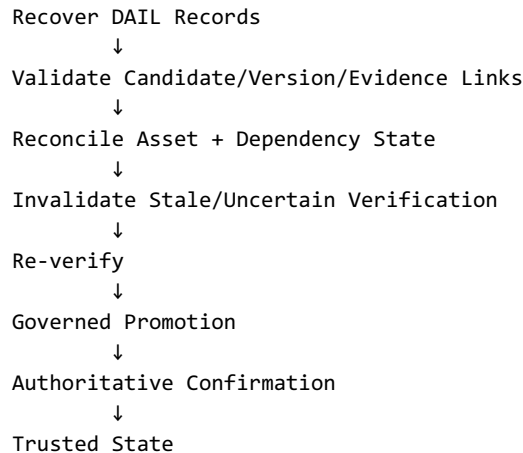
16. AI / LLM Dependency Recovery

- AI is non-authoritative and therefore must not block core security or deterministic recovery.
- If the primary provider is unavailable, queue AI work, use an approved alternate provider, or operate without AI depending on policy.
- Do not automatically switch to an unapproved model/provider during recovery.
- Provider changes must preserve output validation, safety controls, data-minimization rules, and observability.
- AI-generated candidates affected by an outage during generation remain incomplete until validated.
- Model/provider recovery requires safety and evaluation smoke tests before high-impact use.

17. DAIL Recovery

- Recover candidate versions, impact analyses, verification runs, evidence, approvals, and promotion records as a coherent lifecycle set.
- Verification results are never inferred solely from candidate status.
- Any mismatch between candidate version and verification evidence blocks progression.
- Recovered DAIL state is revalidated against authoritative asset/dependency observations.
- Pending verification may be restarted from a clean state when integrity cannot be established.
- Pending promotion is treated as unexecuted unless authoritative evidence proves otherwise.

- Recovered trusted state is not accepted merely because a database record says PROMOTED or TRUSTED.



18. Identity, Secrets, and Key Recovery

- Recovery procedures must preserve least privilege and separation of duties.
- Secrets are restored through managed secret systems, not copied into recovery documents or logs.
- Compromised credentials are rotated rather than restored unchanged.
- KMS/key recovery must be tested independently from application recovery.
- Break-glass access is time-bound, audited, least-privileged, and reviewed after the event.
- Identity-provider outages should support a documented safe degradation mode without weakening tenant isolation.

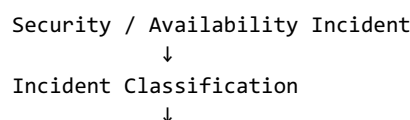
19. Infrastructure and Terraform Recovery

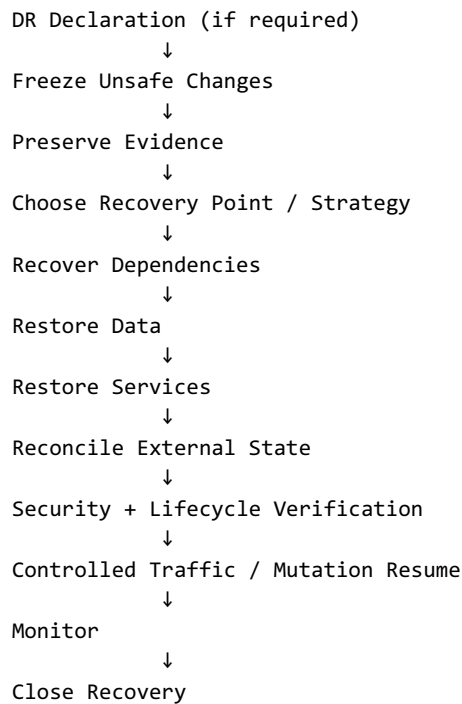
- Terraform state is treated as critical infrastructure metadata and protected independently.
- Recovery uses version-controlled modules and known-good deployment artifacts.
- Terraform plans are reviewed before destructive restoration actions where practical.
- Restored infrastructure is compared with expected architecture and security baselines.
- Drift discovered during recovery becomes explicit reconciliation work.
- Terraform recovery must not destroy evidence or cloud resources merely to force state convergence.

20. Observability and Security Operations During Recovery

- Recovery actions generate normal logs, metrics, traces, audit events, and security events.
- Recovery mode itself is observable through a durable recovery identifier.
- Security operations receives visibility into recovery actions and privileged access.
- Telemetry loss during recovery is treated as a control gap and escalated when material.
- Recovery dashboards show phase, dependency health, RTO elapsed time, RPO point, restore progress, reconciliation backlog, and unresolved uncertainty.
- Recovery evidence remains available after normal operations resume.

21. Incident-to-Recovery Workflow





Disaster recovery and security incident response are coordinated but separate control processes.

22. Failover and Service Restoration

22.1 Failover Preconditions

- Recovery environment is provisioned and security-hardened.
- Required backups/replicas are reachable and integrity-checked.
- DNS/traffic routing controls are available.
- Identity, secrets, and KMS access is operational.
- Database recovery point is identified.
- Application artifacts are available.
- Recovery operators and incident command are assigned.

22.2 Restoration Stages

9. Restore foundational identity, secrets, keys, and network controls.
10. Restore authoritative data and evidence stores.
11. Deploy application/API and validate security controls.
12. Restore workers and queues in controlled capacity.
13. Revalidate cloud integrations in read-only mode.
14. Reconcile pending work and external state.
15. Restore DAIL verification capability.
16. Run security, lifecycle, and smoke tests.
17. Gradually re-enable mutation workflows.
18. Return to normal monitoring and formally close recovery mode.

23. Data Reconciliation and Trusted-State Recovery

Recovery is complete only when distributed state is understood. CloudShield AI must reconcile its internal records with authoritative external systems, especially AWS.

State During Recovery	Required Action
Known Good	May be retained if supporting evidence remains valid.
Stale	Refresh authoritative observation before sensitive progression.
Conflicting	Enter reconciliation; do not choose a side silently.
Unknown	Block consequential progression until resolved or explicitly governed.
Partially Restored	Keep affected workflow degraded/quarantined.
Recovered Candidate	Revalidate exact candidate version and dependencies.
Recovered Trusted State	Reconfirm required promotion evidence and authoritative external state.

Trusted state is earned again when required after a material recovery event; database restoration alone is not sufficient proof.

24. Backup Integrity and Restore Testing

- Backups must be periodically tested for readability, completeness, integrity, and restore usability.
- Restore tests should include database point-in-time recovery, evidence/object recovery, Terraform state recovery, and secret/key recovery.
- Recovery tests must validate tenant isolation and security controls, not only application availability.
- Record restore duration, data-loss point, integrity failures, operator actions, and corrective work.
- Failed restore tests create tracked reliability/security findings.

25. Disaster Recovery Drills

Drill	Frequency Baseline	Purpose
Database Restore	Quarterly	Validate PITR, schema compatibility, integrity and RTO.
Evidence Restore	Quarterly	Validate evidence durability and provenance.
AZ Failure	Quarterly	Validate service rescheduling and recovery.
Regional Recovery	Semiannual	Validate secondary-region strategy where deployed.
Credential/Secret Recovery	Quarterly	Validate secure identity/key restoration.
Ransomware/Destructive Event	Semiannual	Validate isolation, known-good restore, evidence preservation.
Cloud Provider Dependency Outage	Quarterly	Validate degraded mode and queueing.
DAIL Recovery	Quarterly	Validate candidate/version/evidence/verification reconciliation.
Full DR Exercise	Annual	Validate end-to-end command, technical recovery and communications.

26. Recovery Runbooks

- Database corruption / point-in-time restore.
- Object/evidence-store recovery.
- Queue/job replay and reconciliation.
- Application rollback and clean redeploy.
- AWS connection and role-assumption recovery.
- Identity-provider outage.

- Secrets/KMS recovery and credential rotation.
- LLM provider outage / safe degraded mode.
- DAIL lifecycle recovery and verification restart.
- Terraform state/infrastructure recovery.
- AZ failure.
- Regional outage.
- Ransomware/destructive deletion.
- Evidence/audit integrity recovery.

27. Communications and Incident Command

- Declare recovery status, scope, incident commander, technical recovery owner, and security owner.
- Track RTO/RPO decisions and deviations explicitly.
- Communicate confirmed facts separately from hypotheses.
- Record major recovery decisions and approvals.
- Customer communications follow organizational and contractual procedures.
- Recovery closure includes residual risk, unresolved reconciliation items, and follow-up actions.

27.1 Recovery Status

Status	Meaning
ASSESSING	Failure scope and recovery path being determined.
DECLARED	Formal recovery process active.
CONTAINING	Unsafe operations frozen/isolated.
RESTORING	Dependencies/data/services being restored.
RECONCILING	Recovered and external state being compared.
VERIFYING	Security, lifecycle, and service checks running.
RESUMING	Traffic/workflows progressively returning.
STABLE	Normal operation restored; enhanced monitoring remains.
CLOSED	Recovery formally complete; post-recovery actions tracked.

28. Ransomware and Destructive-Event Recovery

- Immediately isolate affected workloads and credentials when compromise is suspected.
- Preserve forensic evidence and avoid overwriting potentially useful artifacts.
- Treat backups as potentially compromised until integrity and provenance are validated.
- Use known-good recovery points with independent protection.
- Rotate compromised credentials before restoring mutation capabilities.
- Restore into isolated infrastructure first.
- Validate application, identity, tenant, security, and evidence controls.
- Re-observe customer cloud state before resuming changes.
- Re-earn trusted state where required.

Never assume that an available backup is safe merely because the backup job succeeded.

29. Regional / Availability-Zone Failure

- MVP should first provide strong single-region, multi-AZ resilience where supported by selected AWS services.

- Regional failover requires separately designed networking, secrets/KMS strategy, database recovery/replication, object replication, DNS/traffic routing, and deployment automation.
- Secondary-region infrastructure must be maintained sufficiently to meet the defined recovery target.
- Customer cloud connections must be revalidated from the recovery region before operations resume.
- Region failover is a material state transition and requires reconciliation of all in-flight jobs and cloud mutations.

30. Control-Plane and Provider Outage

Outage	Required Behavior
CloudShield API	Fail over/restart; preserve durable state; degrade mutation workflows if lifecycle dependencies unavailable.
Database	Restore/fail over; freeze unsafe writes until integrity confirmed.
Object Storage	Queue evidence-dependent work; preserve evidence locally only where explicitly designed.
Queue	Do not blindly replay; use durable job/idempotency records.
LLM Provider	Degrade to deterministic operation, queue AI work, or approved alternate provider.
SIEM/SOAR	Continue native audit/security controls; queue/export later.
External Integration	Queue outbound work; deduplicate/replay inbound events safely.
AWS Customer API	Pause/retry within limits; reconcile ambiguous mutations.
Identity Provider	Use documented safe degradation; never disable tenant isolation.

31. Recovery Security Controls

- Recovery environments must enforce the same or stronger authentication, authorization, tenant isolation, and secret-handling controls as production.
- Recovery operators and break-glass identities are fully audited.
- Backups are encrypted and access-controlled.
- Restored data is not exposed publicly during recovery.
- Recovery artifacts are scanned/validated before deployment.
- Security baselines are checked before traffic or mutation access is enabled.
- Material recovery actions are recorded as audit events and linked to recovery evidence.
- Untrusted or partially restored data is not allowed to influence trusted-state promotion without validation.

32. Recovery Automation Boundaries

Automation	Allowed	Not Allowed
Backup	Automated scheduled backup/replication.	Disabling retention or deleting protected backups without governed authorization.
Restore	Automated restore into controlled recovery environment.	Blindly overwriting authoritative state.
Failover	Automated failover when explicitly designed/tested.	Failing over into an unvalidated insecure environment.
Job Replay	Automated replay of idempotent safe jobs.	Blind replay of uncertain cloud mutations.
Credential Recovery	Automated rotation/reissue through managed systems.	Restoring known-compromised credentials.
AI Recovery	Provider failover to approved configuration.	Switching to unapproved model/provider.
DAIL Recovery	Automated integrity checks/reverification	Marking verification/trust state successful without

	workflows.	evidence.
Promotion	Normal governed promotion after recovery.	Recovery automation directly creating trusted state.

33. Environment-Specific Recovery

Environment	Recovery Policy
Development	Fast rebuild preferred; data can be reconstructed from source/test fixtures; recovery mainly validates automation.
Test	Restore controlled test data and cloud environments; recovery drills can be frequent and destructive.
Staging	Production-like recovery controls and representative DR tests; verify deployment and data migration paths.
Production	Formal incident command, evidence preservation, RTO/RPO tracking, least privilege, reconciliation, and governed resumption.

34. Retention and Recovery Evidence

- Every material recovery event has a recovery ID, declared time, scope, recovery point, operators, and phase history.
- Record backup identifiers, restore operations, hashes/integrity checks, validation results, and reconciliation outcomes.
- Recovery evidence follows security/audit retention policy.
- Recovery logs are not a substitute for durable audit/evidence records.
- Post-recovery reports retain deviations from RTO/RPO and unresolved residual risks.

35. Capacity and Recovery Scaling

- Recovery capacity must be sufficient to support the defined critical workload, not necessarily the maximum normal workload.
- Prioritize Tier 0 and Tier 1 services during constrained recovery.
- Throttle workers to avoid database/API/cloud overload during catch-up.
- Use queue depth and reconciliation backlog to control recovery concurrency.
- Document capacity assumptions used to establish RTO targets.
- Recovery environments must account for AWS service quotas and regional resource limits.

36. MVP Disaster Recovery Baseline

- Multi-AZ deployment for critical regional AWS services where supported and economically justified.
- Automated encrypted database backups with point-in-time recovery.
- Protected/versioned object storage for evidence and artifacts.
- Durable job/idempotency state and controlled replay.
- Version-controlled Terraform and recoverable remote state.
- Managed secrets/KMS recovery procedures.
- Known-good immutable application artifacts.
- Documented application/database/object/worker restore runbooks.
- AWS customer-connection revalidation and post-recovery discovery.
- AI degraded mode and approved provider-failure handling.

- DAIL candidate/version/evidence recovery and re-verification.
- Security operations integration and recovery-specific audit evidence.
- Quarterly restore testing and at least annual end-to-end DR exercise.
- Initial RTO/RPO measurements and tracked improvement backlog.

37. Implementation Sequence

19. Define recovery tiers, business priorities, and validated RTO/RPO targets.
20. Implement database backup/PITR and restore automation.
21. Implement protected object/evidence storage and recovery manifests.
22. Implement durable job/idempotency recovery and safe replay.
23. Automate infrastructure rebuild with Terraform.
24. Implement secrets/KMS recovery procedures and break-glass controls.
25. Create application/API clean-deploy recovery path.
26. Create worker/queue recovery and reconciliation workflows.
27. Implement AWS connection revalidation and post-recovery observation.
28. Implement DAIL recovery integrity checks and re-verification.
29. Implement AI/external dependency degraded modes.
30. Instrument recovery telemetry, audit, dashboards, and metrics.
31. Execute component restore tests.
32. Execute AZ, security/destructive-event, and full DR exercises.
33. Measure results and update RTO/RPO, runbooks, and architecture.

38. Security and Functional Acceptance Criteria

ID	Acceptance Criterion
DR-AC-001	Critical product data has automated, encrypted, recoverable backups.
DR-AC-002	Database recovery can restore to a defined point-in-time and pass domain integrity checks.
DR-AC-003	Evidence/object recovery preserves provenance and integrity metadata.
DR-AC-004	Recovered jobs are idempotency-checked before replay.
DR-AC-005	Unknown cloud mutation outcomes enter reconciliation rather than blind replay.
DR-AC-006	Recovery preserves tenant isolation and authorization controls.
DR-AC-007	Recovery operators and break-glass access are fully auditable.
DR-AC-008	Recovered infrastructure is validated against security baselines before broad service resumption.
DR-AC-009	AI provider outage does not prevent core deterministic security/lifecycle controls from operating.
DR-AC-010	DAIL recovery validates candidate/version/evidence relationships and does not infer verification from status alone.
DR-AC-011	Material recovery events produce durable recovery evidence and audit records.
DR-AC-012	Recovery supports a documented degraded mode when non-critical dependencies are unavailable.
DR-AC-013	RTO/RPO measurements are captured during restore exercises.
DR-AC-014	Restore tests include security, tenant-isolation, integrity, and reconciliation checks.
DR-AC-015	Recovered state cannot directly create TRUSTED state without normal governance, verification, promotion, and authoritative confirmation.

39. Traceability

Upstream Specification	Disaster Recovery Dependency
00 — Master System	Product trust model, end-to-end control loop, product boundary.
01 — SRS	Reliability, resilience, security, data, performance, and operational requirements.
02 — Architecture	Deployment topology, services, storage, queues, trust boundaries.
03 — Functional	Lifecycle behavior, async jobs, failures, recovery workflows.
04 — Product & UX	Degraded states, job status, operator/recovery UX.
05 — Domain & Data	Authoritative data model, evidence, audit, lifecycle records.
06 — State & Lifecycle	State invariants, invalidation, reconciliation, trusted-state rules.
07 — Identity & Dependency	Stable identity, dependency graph, impact and invalidation.
08 — Security & Threat Model	Security controls, threat response, recovery trust boundaries.
09 — Intelligence & AI Architecture	AI abstraction and safe degraded operation.
10 — LLM Integration	Provider failure, routing, retries, usage, provider abstraction.
11 — AI Safety	Safe AI degradation and protection from unsafe recovery behavior.
12–26 — DAIL Package	DAIL state, evidence, verification, testing, experiments, DevOps and implementation.
27 — AWS & Terraform	AWS topology, backups, IAM, Terraform, secrets, observability, DR infrastructure.
28 — API	Safe API recovery, authorization, idempotency, lifecycle-safe commands.
29 — Cloud Integration	AWS reconnection, authoritative observation, mutation reconciliation.
30 — External Integration	External dependency outage, queues, webhooks, replay, integration recovery.
31 — Observability & Logging	Recovery telemetry, audit, evidence, health, alerting.
32 — Security Operations	Incident coordination, containment, evidence, recovery security operations.

40. Engineering Decisions

- Disaster recovery is product-wide and includes DAIL as a recovery-sensitive subsystem.
- Compute is replaceable; authoritative data, evidence, lifecycle state, and recovery metadata require durable protection.
- Recovery must be dependency-ordered and evidence-driven.
- Internal state is never assumed to be authoritative for customer cloud state after a material failure.
- Unknown external outcomes are reconciled rather than blindly retried.
- AI is non-authoritative and must degrade safely during provider outages.
- Recovery automation cannot bypass authorization, DAIL verification, governance, or promotion.
- Trusted state may need to be re-earned after material recovery events.
- MVP prioritizes strong backup/restore, multi-AZ resilience, reconciliation, and tested runbooks before complex multi-region active-active architecture.

41. Definition of Done

- Recovery tiers and validated RTO/RPO targets are documented.
- Automated encrypted database backup/PITR is operational and tested.
- Evidence/object backup and integrity validation are operational.
- Durable job/idempotency recovery and safe replay are implemented.
- Terraform can rebuild required production infrastructure from version-controlled definitions.

- Secrets/KMS recovery and break-glass procedures are tested.
- Application/API and worker recovery paths are documented and automated where practical.
- AWS customer-cloud reconnection and authoritative reconciliation are implemented.
- DAIL recovery validates lifecycle integrity and requires re-verification when appropriate.
- AI and external dependency degraded modes are implemented.
- Recovery telemetry, audit, evidence, dashboards, and alerts are operational.
- Component restore tests and end-to-end DR exercises have passed or have tracked corrective actions.
- Recovery operators cannot bypass governance or directly create trusted state.
- Post-recovery review and improvement workflow is operational.
- Traceability to Specifications 00–32 is maintained.

42. Glossary

Term	Definition
Disaster Recovery (DR)	Processes and technical mechanisms used to restore CloudShield AI after material failure or disaster.
Business Continuity	Ability to continue essential product operations during disruption.
RTO	Target maximum time to restore a capability after a defined failure.
RPO	Target maximum acceptable recoverable data loss measured in time.
Recovery Point	Specific backup/replication point selected for restoration.
Failover	Switching service operation to alternate infrastructure or capacity.
Restore	Reconstructing data or service from protected recovery sources.
Reconciliation	Resolving divergence between recovered internal state and authoritative external state.
Degraded Mode	Restricted operation that preserves safe useful capability during dependency failure.
Break-Glass Access	Emergency access path with strict time, scope, audit, and review controls.
Known-Good State	State supported by validated data, artifacts, security controls, and evidence.
Trusted State	State earned only after required verification, governance, promotion, and authoritative confirmation.
DAIL	CloudShield AI subsystem responsible for dependency-aware lifecycle, impact, verification, and related controls.
PITR	Point-in-time recovery.
Idempotency	Property allowing a retried operation to avoid unintended duplicate side effects.
Recovery Evidence	Durable records proving what was restored, validated, changed, and reconciled.

Final recovery rule: CloudShield AI is not considered safely recovered merely because services are running. Recovery is complete only when critical data, security controls, external state, lifecycle state, evidence, and governance have been validated and reconciled.